SUSTAINABLE DESIGN ENGINEERING submission 1



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Curtis: [redacted]
Leia: [redacted]
Yousuf: [redacted]

Market Research

Purpose: to understand the direction of the aircraft seating industry and the focus of our own design research

Problem

In 2050, aviation emissions will account for **25% of the global carbon budget** [1]

Goal

0% Emission goal by global aviation industry (ICAO) committed Net-Zero [2] Carbon Emissions by 2050

Focus

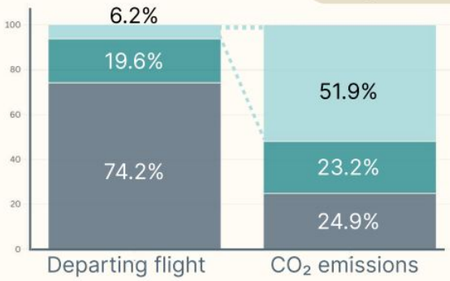
Cabin interiors are **~10% of an aircraft's empty weight** and are replaced **4-5 times** per airframe life [3]

Design scope

Designing for

Long Haul (>4,000km)

CO_2 emissions



Most CO_2 emissions come from long-haul flights, so weight savings there yield the greatest impact. [4]

Design Challenge

Long-haul seats have stricter **design requirements**, making sustainable solutions challenging and a key area for research.

Wide body

Seat Shipset

Wide-body aircraft have largest seatship sets, **maximising mass savings** per flight hour.

CO_2 efficiency

Wide body economy seating is **27% more CO_2 efficient** per passenger-km than narrow body economy, making them optimal for long-haul travel. [5]

Market Trends

In-Flight Entertainment

Modern IFE systems now offer high-definition touch screens and a variety of entertainment options



[6]

Internet Access



[7]

High-speed Wi-Fi and reliable connectivity are now essential to users, prompting airlines to invest in satellite-based internet

Device Accessibility

USB ports and device holders are becoming standard



[8]

MRO system



[9]

With growing focus on Maintenance, Repair, and Overhaul (MRO), airlines are seeking modular, durable and easily swappable seat components

Insights:

Technology for in-flight entertainment and device accessibility is essential for passengers but may hinder seat recyclability

Airlines are focusing on MRO to improve efficiency, which could also support sustainability

The 3 Pillars of Sustainability

Social **Environmental** **Economical**

Our design research should unite the three pillars to meet user, stakeholder, and environmental needs in aircraft seat design.

Market Innovations in Sustainability [10]

Materials	Cost	Weight	Benefit
Bio-Elastomer	✕	−	3D-printable flexible polymer, energy-efficient synthesis
Aluminium	+	−	Proven aviation metal with recycling market projected to reach $101.7 billion by 2032
Bio-composite reinforcements	✕	−	Natural fibre replace carbon fibre
Grape waste leather	+	−	Replaces animal leather and reduces waste
Bio-Plastic	✕	−	Renewable alternative to PP/PET plastics, reduces fossil dependency
Manufacturing			
Bio-inspired design	✕	+	Nature-inspired structures enable high strength-to-weight
Fine 3D-printing metals	✕	+	Zero-waste metal forming up to 20% seat weight reduction
High-performance 3D-printed plastics	−	+	Lightweight polymer composites, recyclable and strong

Insights:

Commercial Insight - Sustainable innovations in aviation remain **costly**, highlighting the need to expand their market or increase airline investment to drive efficiency and lower costs.

Manufacturing Insight - Emerging precision and zero-waste manufacturing methods **reduce material waste**. Our design should anticipate this shift, aligning with future, not just current market capabilities.

Market Insight - In the aviation sector, sustainability initiatives consistently focus on maintaining or reducing weight. This suggests that whatever we do, **overall weight should not be increased** as it is primarily desired in the market.

Next step: consider stakeholder needs to insure seat's commercial viability.

Stakeholders

Purpose: to understand the key stakeholders to identify how sustainability can be implemented feasibly.

Stakeholder Influence & Interest in Aircraft Seat Design

Power

Aircraft manufacturers control certification requirements.

Airline executives set business constraints.

Aviation authorities enforce safety standards.

Airline engineering teams specify features & branding

Seat manufacturers build the seats

Regulatory compliance engineers ensure design meets certification standards.

Passengers' Associations sometimes influence passenger wellbeing policies.

Sustainability Experts assess environmental impact.

Passengers directly experience design.

Cabin crew interact with seats during workflow.

Maintenance & cleaning staff affected by ease of repair and cleanliness.

Interest

Insights:

Design decisions are driven by regulators, manufacturers, and airlines prioritising **compliance, efficiency, and profit.**

Operational staff are underrepresented in early design stages. **Opportunity:** Integrate **crew insights** into the design loop.

Seat manufacturers don't control high-level design decisions, but their **hands-on production experience provides valuable insight.**

Selling Point Considerations[11]

Aligning them with sustainability goals

With Margins under **5%**[12] **small cost shifts can significantly impact profits.**

- **Operational Efficiency:** Streamline workflows
- **Customer Experience:** Focus on comfort, reliability and loyalty
- **Market Positioning:** Define a clear value proposition and brand identity
- **Cost Control:** Manage fuel usage (from excess weight), maintenance, labour, material costs and airport fees
- **Innovation & Technology:** Adopt digital tools for efficiency, predictive maintenance, and enhanced passenger service
- **Regulatory Compliance:** Uphold strict safety and regulatory standards to protect reputation

Insights:

Lighter seat designs and the use of often cheaper **recycled** materials in a **circular system** align **environmental** goals with **cost** control.

Insights:

An **innovative** shift in seat integration is needed to **streamline** airline **operations**, where improved **efficiency** can also drive greater **sustainability**.

Design changes aligned with **brand storytelling** can enhance perceived value for consumers, a **key** driver of **financial success.**

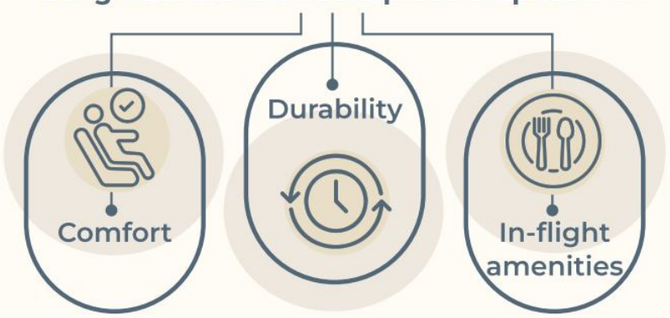
Why Stakeholders are more sustainably inclined[12]

Airlines must disclose carbon footprints, creating reputational pressure.

CORSIA drives greener airline operations via financial and regulatory pressures.

With stakeholders sharing environmental vision, it creates opportunities for **commercially viable sustainable innovation**.

Long-haul stakeholders place emphasis on



Impact of Branding on Seat Lifecycle and Serviceability[10]







- Heavy branding makes spare parts harder to produce and products harder to service.
- Strongly branded cabins reduce the attractiveness of leasing to other operators.
- Seats and furniture cannot be easily resold, limiting their in-service life with secondary operators.
- Branding drives experiential differentiation, and as Elina Kopola, CMF Expert at TrendWorks notes, airlines often purchase seats based on visual appeal.

Opportunities:

- **Modular or removable branding**: Design branding elements that can be swapped or removed
- **Standardised core components**: Keep the underlying seat structure uniform.
- **Integrate sustainability into branding:** Use organic or recycled-looking materials to emphasise circularity's role in the brand.

Next step: explore user experience, a key driver of financial success, to better engage stakeholders.

User Experience

Purpose: identify how sustainable design can support user experience.

Why This Matters?

84% of travellers say their **experience drives loyalty,** [14] **so airlines prioritise it to stay competitive**



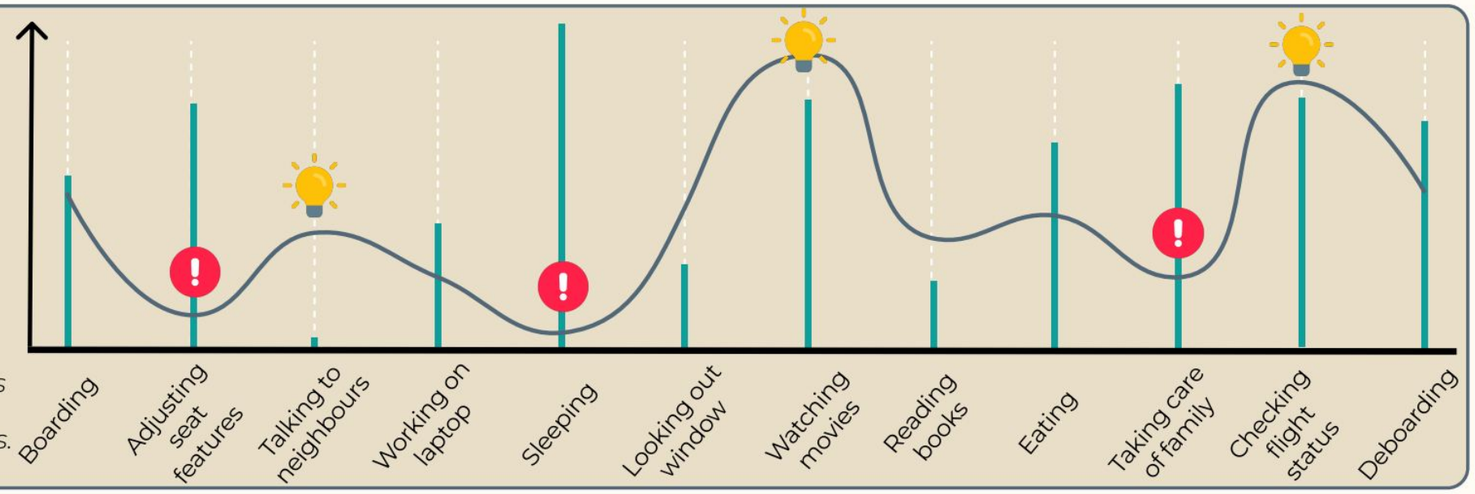
[48]

Experience map

Revealing where importance is high but satisfaction is low.

Satisfaction
Importance

Based on insights from the ERAU passenger satisfaction survey (ages 20–83), representing a broad range of travellers. [15]



Translating ❗Pain Points and 💡oppotunities on Experience Map into Investigation Areas:

❗ Current seats lack effective adjustability.

💡 Low interest in neighbours offers a chance to boost comfort with more privacy.

❗ Disrupted sleep caused by uncomfortable seating is a serious concern.

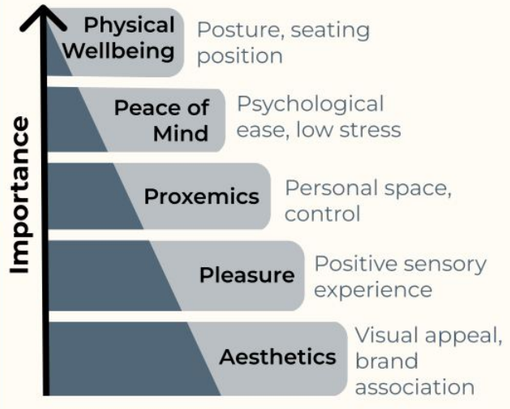
💡 Positive in-flight entertainment use can be leveraged sustainably to enhance passenger experience.

❗ Current seating makes it difficult for passengers to care for family

💡 Passengers' interest in staying informed presents an opportunity to provide updates on environmental impact.

❗ Comfort

Key passenger comfort factors by importance. [16]



Limitation: Able-bodied sample limits generalisability.

Physical wellbeing:
Seats are designed for the 95th percentile male, neglecting other body types. [17]

Peace of mind: [18]
Seat must be designed to withstand 16G.

Proxemics:
Finnair's AirLounge seat reclines in place without reducing rear passenger space. [19]

Pleasure:
Based on study, seat cover material should have low water vapour resistance, thermal insulation and odour retention for best sensory experience. [20]

Safety bias:
Crash-test dummies are male-biased. [21]

Opportunity:
Create **more adjustable seating features** to accommodate diverse body types, improving comfort and safety.



[22]

💡 In-seat inflight entertainment (IFE)

In-seat IFE adds weight, energy use, and e-waste, reducing sustainability.



Opportunity: Digital solutions, like IoT, can replace IFE and simplify maintenance.

💡 Passenger Perceptions of Sustainability

An Ansys survey (*among 6,029 respondents)* found: [23]

63% of consumers worry about aircraft CO_2 emissions

65% would pay more for greener travel

Choosing environmentally friendly options often means sacrificing comfort, time or cost.

Insight: there is a **greater need for costumer awareness** on flight sustainability to make informed trade-offs.



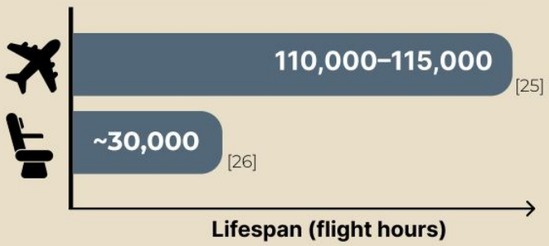
Some airlines display CO_2 emissions in kg, but consumers struggle to interpret the impact of one kilogram. [24]

Opportunity: to design flight search tools that **translate emissions into meaningful personal context.**

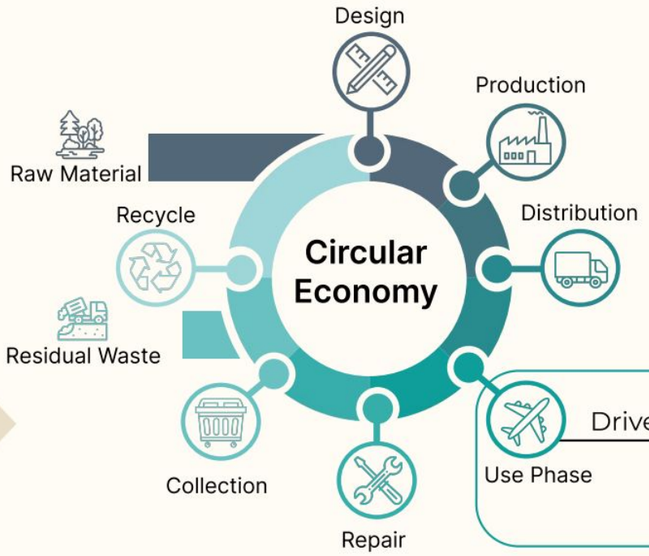
Material Waste

Purpose: To analyse waste and inefficiency in the seat's use phase to drive circular design.

Frequency of seat replacements:



Drives material waste, emphasising need for circular design.



> GREEN CABIN ALLIANCE
> To move away from a linear process of 'take, make and dispose' to a 'waste-free' regenerative process will be the next big impact for our industry.
> **-Elina Kopola**

Advances in disassembly now enable



of componenets to be recycled or reused

Drivers of sustainability issues:

Dominates operational emission ← Fuel consumption | In-flight systems | Cleaning & maintenance → *Directly affects seat lifespan and waste generation*

Environmental impact of Cleaning & Maintanance:

[68] [69] [70]

Material waste



From products and packaging

High water and energy consumption



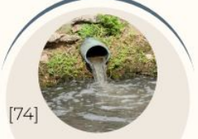
Resource use

Component degradation



Chemicals decrease seat lifespan

Run-off and disposal threatens local ecosystems



Chemical pollution

Routine cleaning at turnaround:

How

Seats are **manually cleaned** via vacuuming, wiping, steam, and chemicals.

If a passenger is sick on a seat, **cushions are completely removed** to access area underneath.

Pain points

Tight spaces (tray tables, armrests, pockets, electronics) lead to low cleaning efficacy and increased labour time.

Cleaning can degrade materials (fabrics, foam, coatings) and compromise fire-retardant properties.

Cleaning each seat consumes **~9,700 litres of fluid per year** [75]

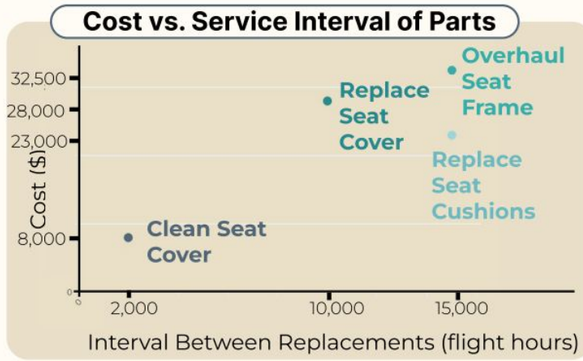
Current Improvements

Shift towards use of **biodegradable and non-toxic cleaning agents**

Use of **antimicrobial coatings** decrease cleaning frequency by 20–50 % [28]

On-site generation of **electrolysed water** lowers transport CO_2 footprint. [29]

Cost vs. Service Interval of Parts



Cumulative Service Cost of Parts



[27]

Purpose:

To examine current airline servicing systems and costs to identify financial burdens and sources of systemic waste.

Assumptions:

- Labour to remove and reinstall fabric covers for cleaning involves 160 man hours
- Seat covers are fabric and dry cleaned

Insights:

Seat covers: incur high costs due to frequent deep cleaning (2,000 FH), often exceeding replacement costs. Can we redesign them to reduce cleaning frequency, cost, and environmental impact?

Seat frames: require costly overhauls every 15,000 FH, as they must be removed from aircraft. Could they be redesigned for easier inspection?

Insights:

Maintenance materials

Poor tracking of them (e.g. chemicals) complicates recycling efforts

Seat material

Materials should be compatible with more advanced cleaning methods

Design for cleaning

Key components should be easily removable for deep cleaning

Next step: explore other stages of the circular economy to maximize our seat's circularity.

Manufacturing and End-Of-Life

Purpose: Analyse the seat's manufacturing and recycling processes to identify sustainability challenges and constraints.

Seat Manufacturing

Problem → Causes → Sustainable Methods → Insights

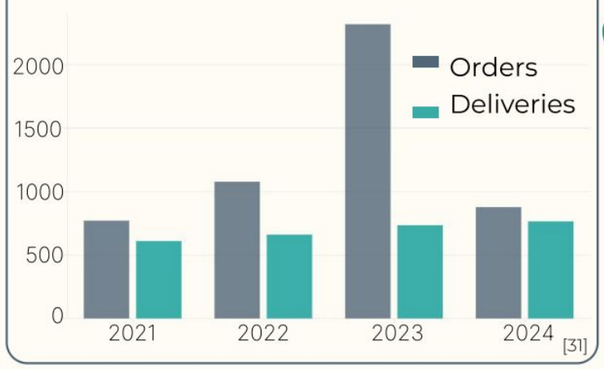
A Gap in the Market...

Slow seat production creates a bottleneck, worsening the gap between aircraft demand and supply.

Impact
Mismatch forces airlines to operate older aircraft for longer.

10–25%
more fuel typically consumed in older planes [30]

Airbus Aircraft Orders vs Deliveries



[31]

> BOEING
> Seat suppliers out there are in shorter capacity. A lot of that is **buyer furnished**, but nevertheless, it holds up an aeroplane
> **-Dave Calhoun**

Because seats are "buyer furnished," Boeing has limited control over the timing and quality of seat deliveries.

Causes of slow seat production [32]

Decentralisation
Aircraft seat components are **sourced from suppliers worldwide**. This is to access global expertise and reduce costs.

Customisation
Many aircraft seats are **still hand-assembled** in production bays rather than mass-produced on moving lines. This is largely due to extensive airline customisation.

Sustainability challenges

Logistical challenges posed by a fragmented supply chain makes it **harder to achieve economies of scale** in sustainable manufacturing. [33]

Sustainability improvements are limited by resource intensive, **low-efficiency manufacturing cycles.** [34]

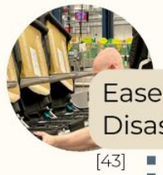
Insight: Customisation and decentralisation increase **complexity** and **production time**; the circular economy could mitigate this through:
- **shared standards**
- **modularity**
- **reuse of seat systems.**

Insight: Seat should be designed for manufacturing **prioritising energy efficiency and intelligent processes.**

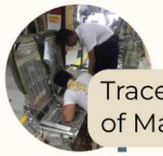
Seat Recycling

Barriers → Solutions → Insight

Key Factors Affecting Recycling


Ease of Disassembly [43]
→ Integrated parts complicate recycling efforts


Material Type [44]
→ Best to maintain material grade within the same industry for quality


Traceability of Material [45]
→ Crucial for quality control and efficient recycling


Transport to Recycling Plants [46]
→ CO_2 emissions from recycling plus transport may exceed emissions saved.


Regulatory Constraints [47]
→ E.g. coatings and fire adhesives impact recyclability.

Real world Solutions

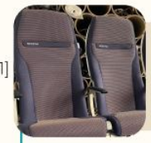

[49] **Designing for disassembly: Loop chair**

All passenger-facing components, from seat cushions to IFE systems, attach to a durable structural frame via mechanical fastenings. The frame remains in the aircraft indefinitely, while each part can be quickly removed, upgraded, repaired, recycled, or reuse.


[50] **Enhancing Traceability: AIRA**

They offer full traceability and certification of materials, showing airlines how interior components can be recycled.

Insight: designing a seat for effective aircraft recycling requires a **holistic systems approach, not just material solutions**.

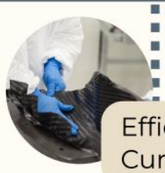

[51] **Regulation-Friendly Recycling: Monova**

The dress cover is made entirely of polyester, including fabric, lamination, and Velcro, eliminating multi-material laminations and bonded layers. This mono-material design enables chemical recycling into new yarns and covers.

Sustainable Manufacturing Methods


[35] Optimised CNC Routing [36]
→ Optimise cutting paths and energy use

Use advanced methods like infrared heating to cure composites


[37] Efficient Curing [38]


[39] JIT Production [40]
→ Manufacture parts only when needed to cut storage energy

Analyse full production process to eliminate non-value-added activities


[41] Value Steam Mapping [42]

Market Benchmarks

Purpose: To identify pain points and best practices within industry benchmarks, building on insights from previous slides.

Seat Selection:

Safran Z400

- Comfort
- Efficiency
- Durability

Selected as the current benchmark of innovation in long-haul economy seating, balancing comfort, weight efficiency and sustainable design.

DOY Ultra Slim Low Carbon Seat

- Sustainability
- Lightness
- Maintainability

Chosen as the epitome of sustainability-driven economy-seat design, recognised with the Red Cabin Trinity Award for Sustainability Concept. [52]

McLaren Racing Seat

- Strength
- Lightness
- Performance

Chosen for its extreme focus on lightness and high-performance durability. It allows us to translate high-performance automotive design principles into aircraft seating

This can be used to identify baseline materials in the current aviation seating market.

This is critical because:

In aviation interiors, **sudden shifts to new material systems are unsustainable** due to established manufacturing lines, recycling streams, certification cycles, and spare-part ecosystems. Rapid changes risk waste, supply-chain disruption, and stranded inventory, so **acknowledging a gradual transition path is important**.

Baseline materials: [62][63][64][65]

- Seat frame: Likely *aluminium alloy*
- Seat shell: Likely *composite-reinforced thermoplastics*
- Cushion: Likely *polyurethane (PU) foam, possibly with memory-foam layer*
- Seat cover: Likely *flame-retardant fabric (polyester blend)*
- Tray table: Likely *polycarbonate or reinforced ABS*
- Fasteners: Likely *stainless steel or aluminium alloy*



[67]

Emirates

$5 billion

program for full cabin interior replacements [66]

This is an example of how **airlines favour large-scale cabin overhauls**, meaning seats can't be replaced individually, making the aviation interiors market inherently spiked.

Sustainability impact: **Causes delays**, extending the use of older aircraft, and **increase waste**.

Insight:

Sustainability may be better achieved through **modular retrofits** to existing seats, rather than full redesigns, enabling gradual market implementation.



[54]

Safran Z400

Features

- Thinner legs improve legroom.
- 8-way adjustable headrest for better neck support.[53]
- Double-deck tray table to combine work and dining.

Pain points

- Fixed backrest/slide-forward mechanism can cause shin clearance issues for taller passengers. [55]
- Narrow layout may compromise comfort.
- Modular design allows customisation but limits personalised seating experience.

Insights

- Design can enhance **perceived space**.
- **Adjustable components** improve comfort and functionality.

Insights

- Ultra-slim structures save weight and space but reduces long-haul comfort, requiring a **balance of ergonomics and sustainability**.
- **Long-term performance of our designed system** must be carefully considered, as unexpected factors could prevent a theoretically repeatable process.

DOY Ultra Slim Low Carbon Seat



[56]

Features

- Composite seat-back with lightweight cushion to reduce weight (only 6.6kg per passenger) [57]
- Primary structure made from recycled aluminium. [58]
- Cushion and dress cover use clip system for easy assembly and disassembly. [59]
- Using recycled materials poses challenges in maintaining performance and meeting strict certifications, yet Doy reports success, claiming its process "can be repeated forever." [58]

Pain points

- Ultra-slim structure reduces padding, making it unsuitable for long-haul comfort.
- Minimalist design restricts customisation and adjustability.



[60]

McLaren Racing Seat

Features

- Flax-fibre composite, instead of carbon fibre, reduces CO_2 footprint by ~75%.[60]
- Continuous shell structure provides high durability and strength.[61]
- Seat is 9% lighter than any equivalent carbon material use and offers significantly better vibration damping [60]

Pain points

- Minimal padding and rigidity for short, high-intensity use.
- Fixed geometry limits customisation for different body types.
- Composite shells can be damaged if mishandled.

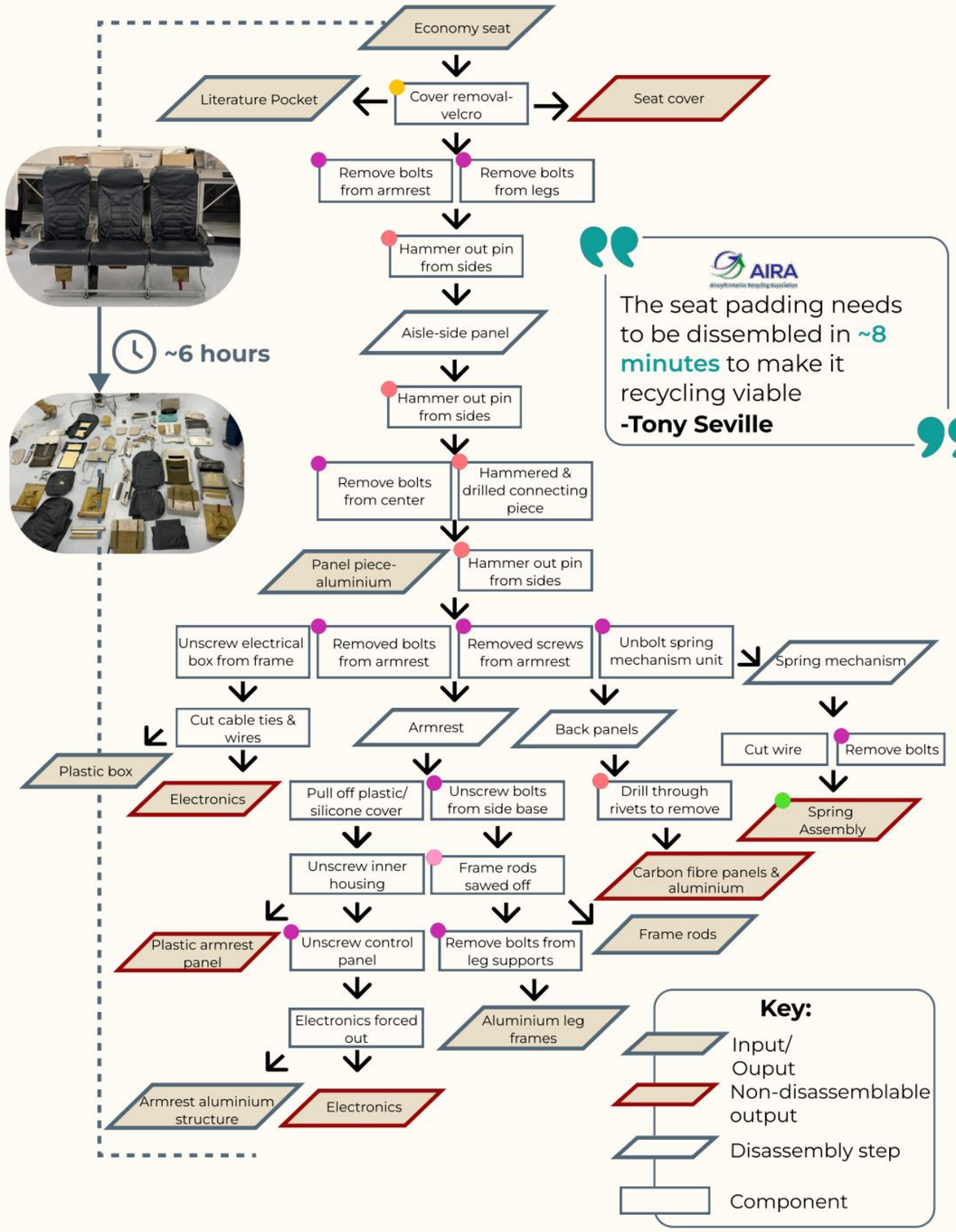
Insights

- **Alternative high-strength composites** cut CO_2 while keeping structural integrity and low weight.
- **Optimised geometry** reduces material use without compromising safety.
- **Lightweight composites are prone to damage**. Maintenance and robustness must be carefully considered for aviation use.

Teardown

Purpose: Evaluating pain points of a seat where full disassembly is not viable

~6 hours

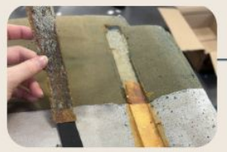


> The seat padding needs to be dissembled in **~8 minutes** to make it recycling viable
> **-Tony Seville**

System wide insight:
The system requires full manual disassembly, making the process labour-intensive. Even padding removal (including Velcro) exceeded the 8-minute target for viable recycling. **Designing fasteners and assemblies for machine-assisted removal** could drastically reduce labour and make recycling scalable and cost-effective.

Padding





- **Seat Cover**

It is the most important part of disassembly as it is most often removed, cleaned, and replaced.

Velcro provides a **fast and efficient fastening method**, making it ideal for quick servicing and improving the cleaning and maintenance experience.

It negatively affects recyclability; even when removed, the adhesive backing **leaves a residue** that prevents the material from being fully recyclable.

Frame







Major pain point during teardown

- **Semi-permanent fasteners (bolts and screws)**

Removal is difficult due to the tight gap between the nut and the frame, which **restricts access**. The unscrewing process is **time-consuming**, and given the large number of fasteners (indicated by the pink dots), this method is not viable.

- **Permanent fasteners (bolts and screws)**

These often needed to be removed using a hammer or drill, which is **time-consuming, requires skill, and often needed two people**. As a result, this method is not also viable for disassembly.

- **Tight-Fit Frame:**

The frame had to be sawn apart as it was **too tightly fitted** within the plastic component. The design did not account for thermal expansion, causing parts to bind together and making disassembly not viable.

Post-Production Modification



- **Unexpected pin**

These additions can reduce **recyclability and complicate disassembly**. For example, removing an unusual pin added to reinforce the spring proved extremely difficult.

Contaminant Accumulation



Dirt

Accumulated dirt and chewing gum **act as contaminants**, reducing recyclability.

Insights:

Fasteners should enable **rapid maintenance and complete disassembly** for recycling.

Frame design **must not obstruct access** to parts that require removal.

Fasteners should be quick and easy to remove and used in smaller quantities, without compromising resistance to vibration and shear stress.

Material joins intended for later separation **should consider thermal expansion** to avoid binding.

Considering a system where **serviceability modifications are planned and communicated** so that recycling teams are aware of them.

Material selection should **consider compression set** to prevent contaminants from entering gaps.

Eco Audit

Purpose: to quantify sustainability success of Doy's Ulta-Slim seat

Assumptions & Limitations

DOY's Ultra-Slim Seat



Disclaimer:
This is a concept design, not the one exhibited at the workshop; manufacturing techniques are therefore uncertain, so assumed to follow industry standards.

Description

One row of **short-haul** aircraft seats with a **five-year service life** (warranty expiry). Seats are **made** from fully **recycled aluminium** and **assumed 100% recyclable at end of life**.

Eco Audit Limitations

- Impact on **human health**.
- **Social ethics** of **offshoring**.
- **Economical impacts** for **exporting countries.**
- Depletion of **natural resources.**

1. Manufacturing Assumptions

- Metals **--> if solid or shaped (forging), --> if Hollow (rolling), --> if Fasteners (wire drawing),**
- Plastics **--> if joints (extrusion), --> else Moulded**

Unspecified manufacturing processes were assumed to **follow standard aerospace practices.**

2. Materials Assumptions

- Polycarbonate **--> Polycarbonate unfilled, flame retarded**
- Carbon Fibre **--> Carbon Fiber 50%**
- Polyoxymethylene **--> Polyoxymethylene unfilled copolymer**
- Steel **--> Low Alloy Steel AISI 4130, Quenched & Tempered**

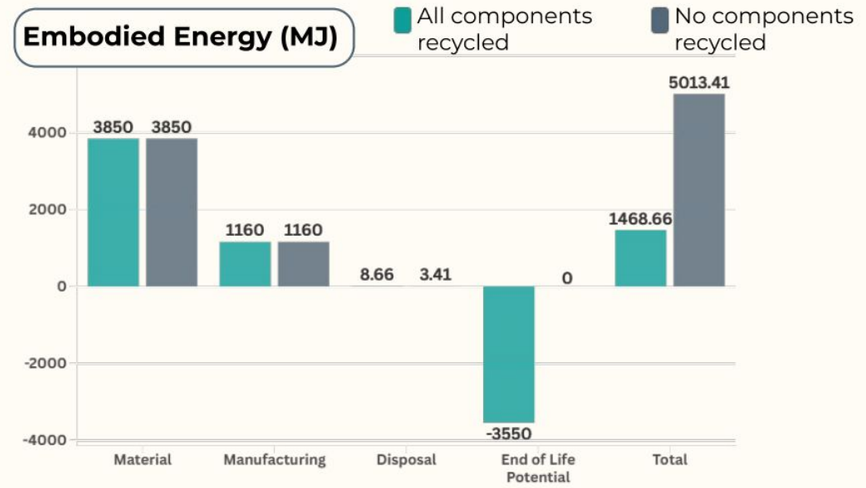
Unspecified materials were assumed to be **typical aerospace-grade** materials, though **some remained undetermined** despite research.

AIRA

Although many assumptions were necessary due to reliance on public data, DOY's designs benefit from full partnerships with AIRA, providing greater transparency and making 100% recyclability far more feasible.

How recycleability can reduce embodied energy

Aim: to quantify design-for-recycling benefits.



Since **99.8%** of **embodied energy** comes from operational fuel use (an unavoidable consequence of aviation) it was excluded from analysis to enable clearer comparison of other contributing factors.

Recycling all components of the seat could reduce its **total embodied energy** by up to **70.7%**.

Insight: The **system is never perfect** (total embodied energy under recycling is still 1,468.66 MJ), so **balancing material longevity with recyclability is key**.

Comparing Embodied Energy with Tear Down Design

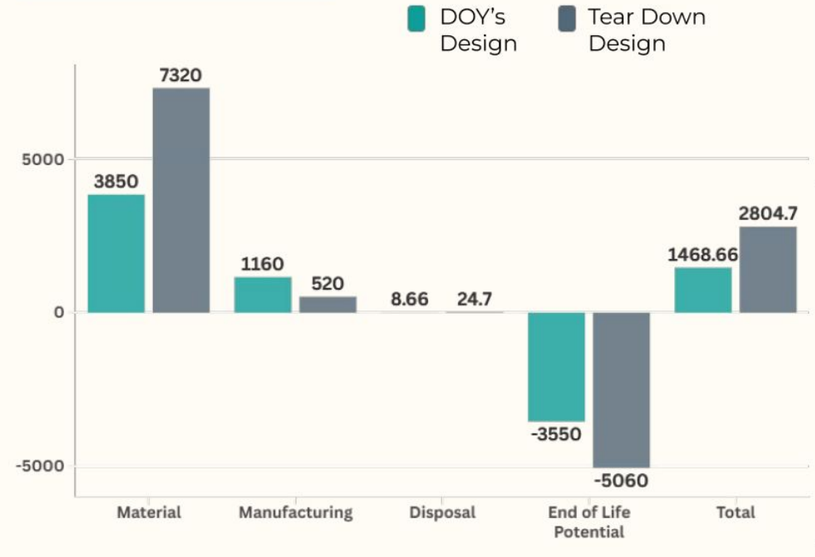
Aim: To comparatively illustrate the embodied energy savings achieved through Doy's model

Functional Unit:

DOY' Design:
One row of short-haul aircraft seats (3 seats) with a five-year service life (warranty expiry), based on DOY's ultra-slim design.

Tear-Down Seat:
Functional Unit: One row of short-haul aircraft seats (3 seats) with a seven-year service life, based on the teardown design.

Embodied Energy (MJ)



- The **teardown seat** shows significantly **higher embodied energy during the material phase**.
- Its **better end-of-life potential is associated to its large proportion of mass being aluminium, which is highly recyclable** (29.5kg vs 6.99kg per row).
- It retains nearly **twice the total embodied energy** than DOY's design by the end of the life cycle.

Teardown seat assumptions:
(limiting graph's accuracy)

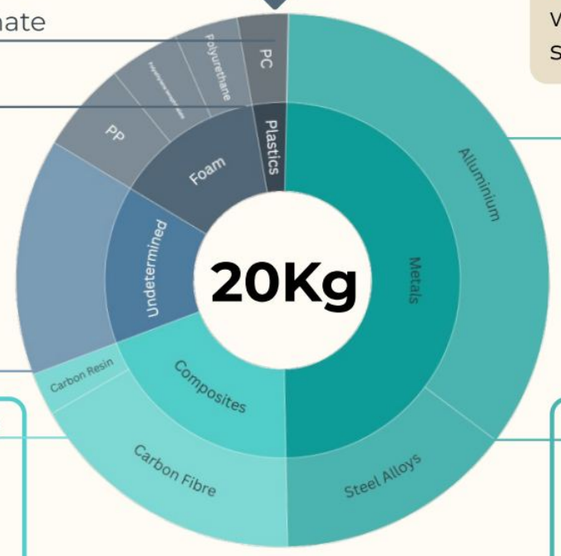
- Only the aluminium parts of seat is recyclable.
- The material composition was based on inferred assumptions without inclusion of material grades as only the seat covers were labelled with material information.

Insight:
The teardown seat's higher mass (42kg vs 20kg per row) drives its greater total embodied energy, despite its aluminium content contributing to end-of-life potential, highlighting that **reducing material mass is also very applicable in lowering seat design's embodied energy.**

Doy Seat Materials

Purpose: Breaking down materials to assess the viability of the recyclability of claimed recyclable components.

Material Weight Distribution



> DOY DESIGN
>
> All the components are recyclable, but the three layers in the cushioning make it really complicated to separate them. In theory, though, it's recyclable.
> **-Gary Doy**

Insight
Although a material may be cited as technically recyclable, **recycling may not be viable**. To ensure true recyclability, the focus should be on widely used industrial materials like aluminium or create recycling systems alongside design.

Aluminium

The majority of the seat's weight distribution is aluminium, which constitutes the full frame because of its favourable strength-to-weight ratio and structural efficiency. This is sensible given its **widespread industrial use.** Its classified by the EU and US as a critical material with potential supply risks.

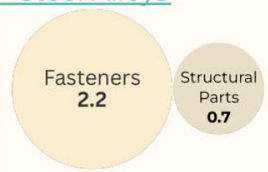
Undetermined Materials

Undetermined materials are those not specified in the BOM, as mentioned earlier. They account for over 10% of the seat's total weight and **cannot be recycled due to the lack of information.**

Carbon Fibre

Studies show **carbon fibre is mostly landfilled**, as recycling and incineration are costly, emit high GHGs, and have low market demand.

Steel Alloys



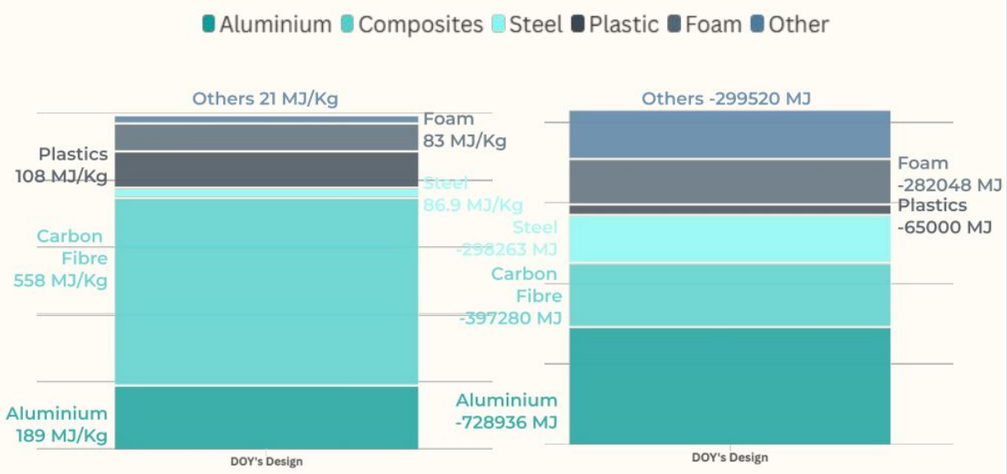
Fasteners represent ~10% of a seat's total mass.

Insights
A fastener-less design can reduce seat weight, yielding substantial long-term impact.

Materials Embodied Energy per Kg

Carbon fibre has very high embodied energy per kilogram and comparatively low end-of-life potential.

Embodied Energy (MJ/Kg) **End of Life Potential (EoL)**



Insight: Using carbon fibre offers high durability and low weight, but comes with the **trade-off of higher total embodied energy** in the seat

Aluminium Circularity Flow Chart

Aluminium circularity was analysed because it is the main component in the seat design (~35%) and has established recycling processes in the industry plus is one of the main contributor materials for embodied energy.

Artificial

Aluminium 7075-T651 is a engineered alloy, not a naturally occurring material, so its production is more complex.

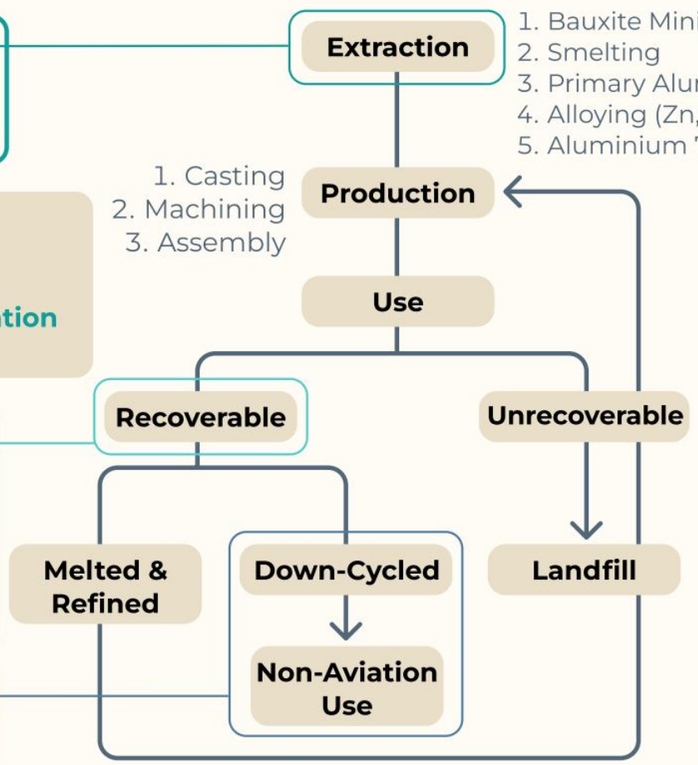
Insight:
The extraction process is costly, creating a market incentive for sustainable practices. Other material recycling may be less viable due to **limited innovation incentives** rather than technical impossibility.

Recycled & Reused

During recycling, only about 35% of Aluminium 7075 can be restored to its original grade, while the remainder is typically downcycled into lower-strength alloys, often from the 3000 series. [76]

Downcycling Industries

Aluminium 3000 series are used for general products like soft drinks cans [77]



HMW page

Purpose: to translate insights into "How Might We" questions to guide design research.

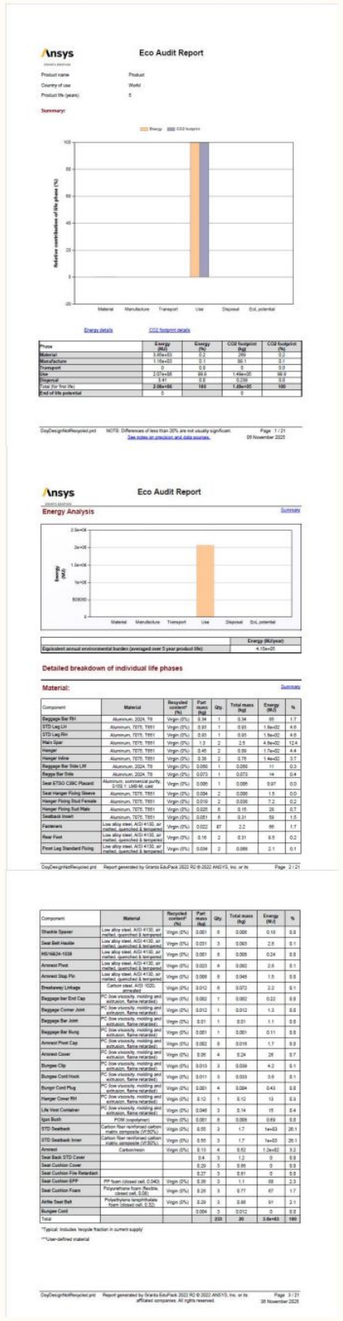
Source	Insights and requirements	HMW
Stakeholders page Stakeholders page Material waste page	**Heavy branding reduces serviceability and limits customisation**, leading to negative economic and environmental impacts. **Brand differentiation remains essential** for passenger experience and stakeholder value. **Seat covers** drive high costs and hinder circularity due to need for **frequent cleaning and replacement**, highlighting need for easy cleaning processes and recyclability.	**How might we** redesign long-haul economy seats with modular branding interfaces, for enhanced user perceived value, which can be quickly cleaned or refreshed and operate within a fully circular system? **Leia**
Stakeholders page User experience page Market benchmark page Doy Seat Materials	The primary market driver is **reducing weight** to control costs and lower CO_2 emissions. Seat frames must withstand 16 G loads. **High-strength composites are the leading innovation** for balancing lightness with structural integrity. **Current composite** use is **not viably recyclable** and is therefore most often sent to landfill.	**How might we** design a system for airplane seat composite parts to be repurposed into other viable systems after airline service life to improve the circular business model?
Teardown page Manufactoring and end-of-life page Stakeholders page Material waste page	**Planned service modifications** help **not create confusion** for recycling teams. **Overly integrated parts complicate recycling** and end-of-life processing. **Operational staff are often under represented** in early design stages. **Extending seat lifespan through easy serviceability is vital** to reduce material waste from replacements.	**How might we** enable modular upgrades in airline-specific long-haul economy seats to prolong their lifespan, allowing components to be quickly refreshed and serviced, and keeping end-of-life recycling straightforward?
Doy Seat Materials Material waste page Teardown page	**A fastener-less approach** could cut seat weight and deliver lasting fuel efficiency gains. **Easier seat removal** would reduce costly overhaul time and simplify inspections. Joints designed for disassembly must account for **thermal expansion** to prevent binding.	**How might we** rethink seat-to-aircraft fastening to enable simple, individual seat removal during overhauls, streamline worker efficiency, and minimise fastener weight to improve fuel performance? **Curtis**
Stakeholders page Market benchmark page	**Emerging market innovations** are making materials and manufacturing more sustainable so our design should align with future capabilities. Sustainability should enable **gradual market adoption** to avoid the environmental impact of sudden large-scale overhauls.	**How might we** design a reconfigurable seat base that allows airlines to implement new sustainable materials gradually without grounding fleets or disrupting maintenance, enabling scalability of the seat over time? **Karanjeet**

Appendix

1] Alfaro VN, Chankov S. The perceived value of environmental sustainability for consumers in the air travel industry: A choice-based conjoint analysis. Journal of Cleaner Production [Internet]. 2022 Nov 2;380:134936. Available from: https://www.sciencedirect.com/science/article/pii/S0959652622045097
2] ICAO Global Framework for SAF, LCAF and other Aviation Cleaner Energies [Internet]. Available from: https://www.icao.int/sites/default/files/sp-files/environmental-protection/GFAAF/Documents/ICAO-Global-Framework-on-Aviation-Cleaner-Energies_24Nov2023.pdf
3] Graver B, Rutherford D, Zheng S. OCTOBER 2020 CO 2 EMISSIONS FROM COMMERCIAL AVIATION [Internet]. 2020. Available from: https://theicct.org/wp-content/uploads/2021/06/CO2-commercial-aviation-oct2020.pdf
4] Chart of the week: CO2 output by flight length [Internet]. TNMT. Available from: https://tnmt.com/newsletter-snippets/chart-of-the-week-co2-output/
5] Graver B, Rutherford D, Zheng S. OCTOBER 2020 CO 2 EMISSIONS FROM COMMERCIAL AVIATION [Internet]. 2020. Available from: https://theicct.org/wp-content/uploads/2021/06/CO2-commercial-aviation-oct2020.pdf
6] r/Canada [Internet]. Reddit.com. 2025. Available from: https://www.reddit.com/r/aircanada/comments/16q5umg/air_canada_inflight_entertainment_system/?rdt=54853
7]White MJ. Is In-Flight Wi-Fi Worth The Cost Or Should You Save Your Money? [Internet]. SlashGear. 2023 [cited 2025 Nov 13]. Available from: https://www.slashgear.com/1219187/is-in-flight-wi-fi-worth-the-cost-or-should-you-save-your-money/
8] Hafiz Johari, commercial jets charging port. Shutterstock. Available from: https://www.shutterstock.com/search/commercial-jets-charging-ports
9] Aero Care BD [Internet]. Aero Care BD. 2017 [cited 2025 Nov 13]. Available from: https://aerocarebd.com/
10] SUSTAINABLE CABIN DESIGN New Approaches in Sustainable Aircraft Interior Design [Internet]. 2022. Available from: https://www.ati.org.uk/wp-content/uploads/2022/03/FZO-AIR-POS-0039-Sustainable-Cabin-Design.pdf
11] Ten Proven Strategies for Airline Profitability and Success [Internet]. Aviaproconsulting.com. 2024. Available from: https://aviaproconsulting.com/article/post/ten-proven-strategies-for-airline-profitability-and-success
12] Mitchell A. Why Future Airplane Seats Will Be Made From Bio Plastics [Internet]. Simple Flying. 2025 [cited 2025 Nov 13]. Available from: https://simpleflying.com/future-airplane-seats-bioplastics/
13] Aviation Fuels (SAF) [Internet]. Icao.int. 2025. Available from: https://www.icao.int/SAF
14] Customer Experience In The Airline Industry Statistics: Market Data Report 2025 [Internet]. Gitnux.org. 2025. Available from: https://gitnux.org/customer-experience-in-the-airline-industry-statistics/
15] Scholarly Commons | Embry-Riddle Aeronautical University Research [Internet]. commons.erau.edu. Available from: https://commons.erau.edu
16] Ahmadpour N, Robert JM, Lindgaard G. Aircraft passenger comfort experience: Underlying factors and differentiation from discomfort. Applied Ergonomics. 2016 Jan;52:301–8. Available from: https://pubmed.ncbi.nlm.nih.gov/26360222/
17]file:///C:/Users/curti/OneDrive/Desktop/AC_25_785-1B.pdf
18] State of Global Aviation Safety ICAO Safety Report 2019 Edition [Internet]. Available from: https://www.icao.int/sites/default/files/sp-files/safety/Documents/ICAO_SR_2019_final_web.pdf
19] Jones R. ANA is the 2nd airline to install no-recline business class seats – and they look great [Internet]. Head for Points. 2025 [cited 2025 Nov 13]. Available from: https://www.headforpoints.com/2025/06/30/ana-no-recline-business-class-seat/comment-page-3/
20] Bartels VT. Thermal comfort of aeroplane seats: influence of different seat materials and the use of laboratory test methods. Applied Ergonomics. 2003 Jul;34(4):393–9. Available from: https://www.sciencedirect.com/science/article/abs/pii/S0003687003000589
21] Eden P. How airline seats are tested [Internet]. Aerospace Testing International. 2025 [cited 2025 Nov 13]. Available from: https://www.aerospacetestinginternational.com/features/how-airline-seats-are-tested.html
22] JAMCO Group Obtains FAA Approval for Dynamic Aircraft Seat Testing Facility | JAMCO News | News about JAMCO | JAMCO Corporation [Internet]. Jamco.co.jp. 2017 [cited 2025 Nov 13]. Available from: https://www.jamco.co.jp/en/news/news/news3287806656275075116.html
23] Ansys. Ansys Study Finds More than 60% of Consumers are Concerned About CO2 Emissions [Internet]. Prnewswire.com. Cision PR Newswire; 2023. Available from: https://www.prnewswire.com/news-releases/ansys-study-finds-more-than-60-of-consumers-are-concerned-about-co2-emissions-301853854.html
24] Alfaro VN, Chankov S. The perceived value of environmental sustainability for consumers in the air travel industry: A choice-based conjoint analysis. Journal of Cleaner Production [Internet]. 2022 Nov 2;380:134936. Available from: https://www.sciencedirect.com/science/article/pii/S0959652622045097
25] aviator. How Long Does a Commercial Aircraft Last? - aviationfile [Internet]. aviation related posts, aviation pioneers and aviation accidents. 2023. Available from: https://www.aviationfile.com/how-long-does-a-commercial-aircraft-last/
26] StreetInsider.com - Reuters [Internet]. StreetInsider.com. 2021 [cited 2025 Nov 13]. Available from: https://www.streetinsider.com/Reuters/Inside%2Bthe%2Bairline%2Bseat%2Bindustry%2Bcrisis%2Bdelaying%2Bjet%2Bdeliveries
27] Aircraft Commerce. Costs of narrowbody interior refurbishment [Internet]. Issue 68. London (UK): Aircraft Commerce Ltd; 2010 Feb/Mar [cited 2025 Nov 13]. Available from: https://www.aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs/Maintenance/2010/ISSUE68_MTCE_B.pdf
28] TeamAviation [Internet]. Teamaviation.in. 2020 [cited 2025 Nov 13]. Available from: https://teamaviation.in/post/advancements-in-aircraft-cleaning-supplies-ensuring-hygiene-and-safety-3627
29] Electrolyzed Water for Cleaning & Disinfection - What is it? [Internet]. New HOCLHUB. 2025 [cited 2025 Nov 13]. Available from: https://hoclhub.com/electrolyzed-water-for-cleaning-disinfection-what-is-it/
30] Field J. The Life of a Seat - Three Sixty Degrees [Internet]. Three Sixty Degrees. 2024 [cited 2025 Nov 13]. Available from: https://threesixtydegrees.net/the-life-of-a-seat/
31] Airbus. Orders and Deliveries | Airbus [Internet]. Airbus. 2024. Available from: https://www.airbus.com/en/products-services/commercial-aircraft/orders-and-deliveries
32] Hepher T. Inside the airline seat industry crisis delaying jet deliveries. Reuters [Internet]. 2025 Jul 2; Available from: https://www.reuters.com/business/aerospace-defense/inside-airline-seat-industry-crisis-delaying-jet-deliveries-2025-07-02/
33] Overcoming Supply Chain Fragmentation: An In-depth Guide | Cargologik [Internet]. www.cargologik.com. Available from: https://www.cargologik.com/blog-post/overcoming-fragmentation-in-supply-chain-systems-a-comprehensive-guide
34] Hepher T. Inside the airline seat industry crisis delaying jet deliveries. Reuters [Internet]. 2025 Jul 2; Available from: https://www.reuters.com/business/aerospace-defense/inside-airline-seat-industry-crisis-delaying-jet-deliveries-2025-07-02/
35] Fitch C. Mill Brass & Aluminum with a CNC Router | Popular Woodworking [Internet]. Popular Woodworking. 2021 [cited 2025 Nov 13]. Available from: https://www.popularwoodworking.com/techniques/mill-brass-aluminum-with-a-cnc-router/
36] badachou91@gmail.com. Towards Greener Manufacturing: Advancing Sustainable CNC Machining Practices [Internet]. TonZa Making. 2025. Available from: https://tonzamaking.com/blog/towards-greener-manufacturing-advancing-sustainable-cnc-machining-practices/
37] bercella-laura. Bercella - first steps of carbon fiber layup process [Internet]. Bercella. 2021. Available from: https://bercella.it/en/stories-en/carbon-fiber-layup/
38] Infrared heat processes and UV curing for composite materials [Internet]. Noblelight.com. 2024 [cited 2025 Nov 13]. Available from: https://www.noblelight.com/en/etc/industries_and_applications/uv_ir_af_applications/infrared_heat_processes_and_uv_curing_for_composite_materials.html
39] AV Birch [Internet]. AV Birch. 2025 [cited 2025 Nov 13]. Available from: https://www.avbirch.co.uk/
40] DataCalculus. Implementing Just-in-Time Manufacturing in Aviation [Internet]. DataCalculus. 2025 [cited 2025 Nov 13]. Available from: https://datacalculus.com/en/blog/aviation-and-aerospace-component-manufacturing/process-engineer/implementing-just-in-time-manufacturing-in-aviation
41] Recaro Aircraft Seating streamlines final assembly line - Aerospace Manufacturing and Design
42] Salwin M, Pszczółkowska K, Pałęga M, Kraslawski A. Value-Stream Mapping as a Tool to Improve Production and Energy Consumption: A Case Study of a Manufacturer of Industrial Hand Tools. Energies [Internet]. 2023 Jan 1;16(21):7292. Available from: https://www.mdpi.com/1996-1073/16/21/7292
43] Griff Z. How airplane seats are actually made - The Points Guy [Internet]. The Points Guy. 2023. Available from: https://thepointsguy.com/airline/how-airplane-seats-are-made/
44] Online Fabric & Haberdashery Shop» Myfabrics.co.uk [Internet]. Myfabrics.co.uk. 2025 [cited 2025 Nov 13]. Available from: https://www.myfabrics.co.uk/?utm_campaign=BI_UK_Stoffe_Generisch&utm_medium=CPC&utm_source=bing&utm_term=%2Buk%20%2Bfabrics&msclkid=80d2a3cc898c18232e34cac37716db5b&utm_source=bing&utm_medium=cpc&utm_campaign=BI_UK_Stoffe_Generisch&utm_term=%2Buk%20%2Bfabrics&utm_content=Stoffe_Allgemein
45] Cabin Seat Repair, Refurbishment and Overhaul – CHARLES AEROSPACE INDUSTRIES, INC. [Internet]. Available from: https://charles-aeroindustries.com/cabin-seat-repair-refurbishment-and-overhaul/
46] Mufson S. Global carbon dioxide emissions hit new highs last year, says IEA report. Washington Post [Internet]. 2023 Mar 2; Available from: https://www.washingtonpost.com/climate-environment/2023/03/02/emissions-hit-carbon-dioxide/
47] McGinnis C. Airlines, hotels to implement “electrostatic spraying” vs. coronavirus [Internet]. SFGATE. 2020 [cited 2025 Nov 13]. Available from: https://www.sfgate.com/travel/article/Airlines-hotels-cleaning-guns-15229928.php
48] Expedia. Book Cheap Holidays: Flights, Hotels, Car Hire | Expedia [Internet]. Expedia.co.uk. 2024. Available from: https://www.expedia.co.uk/
49] Loop Chair Concept from Studio ID Champions Circular Design - APEX [Internet]. APEX. 2024 [cited 2025 Nov 13]. Available from: https://apex.aero/articles/loop-chair-concept-from-studio-id-champions-circular-design/
50] max2testbase. Recycling Aircraft Interiors - Aircraft Interior Recycling Association [Internet]. Aircraft Interior Recycling Association. 2020. Available from: https://www.aira-international.com/recyclingaircraftinteriors/
51] Elina Kopola [Internet]. Trendworks.co.uk. 2025 [cited 2025 Nov 13]. Available from: https://trendworks.co.uk/
52] Gavine A. The winners of the RedCabin Trinity Awards are announced [Internet]. Aircraft Interiors International. 2024 [cited 2025 Nov 13]. Available from: https://www.aircraftinteriorsinternational.com/news/events-and-awards/the-winners-of-the-redcabin-trinity-awards-are-announced.html?utm_source
53] » SAFRAN Z400 Seat by Safran [Internet]. Thedesignsoc.com. 2021 [cited 2025 Nov 13]. Available from: https://thedesignsoc.com/z400-seat-safran/
54] Economy class seat Z400 [Internet]. Safran. 2023. Available from: https://www.safran-group.com/products-services/economy-class-seat-z400
55] Alkema B. Behind the scenes with Safran’s secret Signature no-recline seat [Internet]. Runway Girl. 2024. Available from: https://runwaygirlnetwork.com/2024/06/safrans-signature-no-recline-economy-seat/
56] Alkema B. Press Release: Heart fits DOY Design Ultra-Slim seats to demonstrator [Internet]. Runway Girl. 2024 [cited 2025 Nov 13]. Available from: https://runwaygirlnetwork.com/2024/09/doy-design-ultra-slim-seats-heart-aerospace/
57] Seat US. DOYDESIGN [Internet]. DOYDESIGN. 2023 [cited 2025 Nov 13]. Available from: https://www.doydesign.com/news/ultra-slim-seat-aix-2023
58] DOYDESIGN [Internet]. DOYDESIGN. 2023. Available from: https://www.doydesign.com/news/recycle-and-reuse
59] Heart. DOYDESIGN [Internet]. DOYDESIGN. 2024. Available from: https://www.doydesign.com/news/heart-aerospace-select-doy-ultra-slim-seats
60] McLaren Racing - Revealed: How McLaren is pioneering the use of sustainable composites in F1 [Internet]. www.mclaren.com. 2020. Available from: https://www.mclaren.com/racing/sustainability/natural-fibre-sustainable-composite-racing-seat/
61] Butcher L. McLaren harnesses natural composite fibers [Internet]. Professional Motorsport World. 2020. Available from: https://www.pmw-magazine.com/news/materials/mclaren-harnesses-natural-composite-fibers.html
62] Hasan Y, Hasan I, Aliabadi AA, Bahram Gharabaghi. Comparative Life Cycle Assessment (LCA) in the Aerospace Industry Regarding Aviation Seat Frame Options. Sustainability [Internet]. 2025 Apr 3;17(7):3188–8. Available from: https://www.mdpi.com/2071-1050/17/7/3188
63] Pierobon M. Under construction: Composite materials replace traditional aircraft materials - Aviation Business News [Internet]. Aviation Business News. 2019 [cited 2025 Nov 13]. Available from: https://www.aviationbusinessnews.com/cabin/composite-materials-aircraft
64] Matthews S. Aircraft Seat Material [Internet]. Scribd. 2025 [cited 2025 Nov 13]. Available from: https://www.scribd.com/document/255997715/Aircraft-Seat-Material
65] SKY airline seat – Orelio [Internet]. Orelio.com. 2021 [cited 2025 Nov 13]. Available from: https://orelio.com/project/lightweight-airline-seat
66] Emirates To Spend 5 Billion On Aircraft Upgrades - Dj’s Aviation [Internet]. Dj’s Aviation. 2025 [cited 2025 Nov 13]. Available from: https://djsaviation.net/emirates-to-spend-5-billion-on-aircraft-upgrades/
67] PlusPNG. Emirates PNG - PlusPNG [Internet]. PlusPNG. 2017 [cited 2025 Nov 13]. Available from: https://pluspng.com/emirates-png-6664.html
98] Sustainable Cabin [Internet]. Iata.org. 2025. Available from: https://www.iata.org/en/programs/sustainability/cabin-waste
69] Aircraft NRC (US) C on AQ in PC of C. Chemical Contaminants and Their Sources [Internet]. www.ncbi.nlm.nih.gov. National Academies Press (US); 2002. Available from: https://www.ncbi.nlm.nih.gov/books/NBK207480/
70] Sustainable Aviation Maintenance Practices: Green Operations - Clarity Airframe Blog [Internet]. Clarity Airframe. 2024 [cited 2025 Nov 13]. Available from: https://clarityairframe.com/blog/sustainable-aviation-maintenance-practices/
71] Detergents Pallet | QUICK LIQUIDATIONS PALLETS INC. [Internet]. QUICK LIQUIDATIONS PALLETS INC. 2025 [cited 2025 Nov 13]. Available from: https://quickliquidationspallets.com/product/detergents-pallet/
72] Detergents Pallet | QUICK LIQUIDATIONS PALLETS INC. [Internet]. QUICK LIQUIDATIONS PALLETS INC. 2025 [cited 2025 Nov 13]. Available from: https://quickliquidationspallets.com/product/detergents-pallet/
73] Investing in the Future of Sustainable Luxury: Tapestry [Internet]. 3blmedia.com. 2025 [cited 2025 Nov 13]. Available from: https://www.3blmedia.com/news/investing-future-sustainable-luxury-tapestry-expands-partnership-gen-phoenix
74] Wright O. Yorkshire Water invests £445k to cut Northallerton storm overflow. BBC News [Internet]. 2025 Jan 17; Available from: https://www.bbc.co.uk/news/articles/czdl43gyn79o
75] Field J. The Life of a Seat - Three Sixty Degrees [Internet]. Three Sixty Degrees. 2024. Available from: https://threesixtydegrees.net/the-life-of-a-seat/
76] 7075-T651 Aluminum: Properties and Applications [Internet]. Howard Precision Metals. 2018. Available from: https://www.howardprecision.com/7075-t651-aluminum-properties-and-applications/
77] Consorzio Imballaggi Alluminio (CiAl). Aluminium Packaging – Guidelines for an Eco-Friendly Design [Internet]. Milan (Italy): CiAl; 2018 Nov [cited 2025 Nov 13]. Available from: https://www.cial.it/wp-content/uploads/2021/04/Aluminium_Packaging_Guidelines_for_an_ecofriendly_design_CIAL.pdf
Icons] Flaticon. Flaticon, The Largest Database of Free Vector Icons [Internet]. Flaticon. 2025. Available from: https://www.flaticon.com/



Appendix

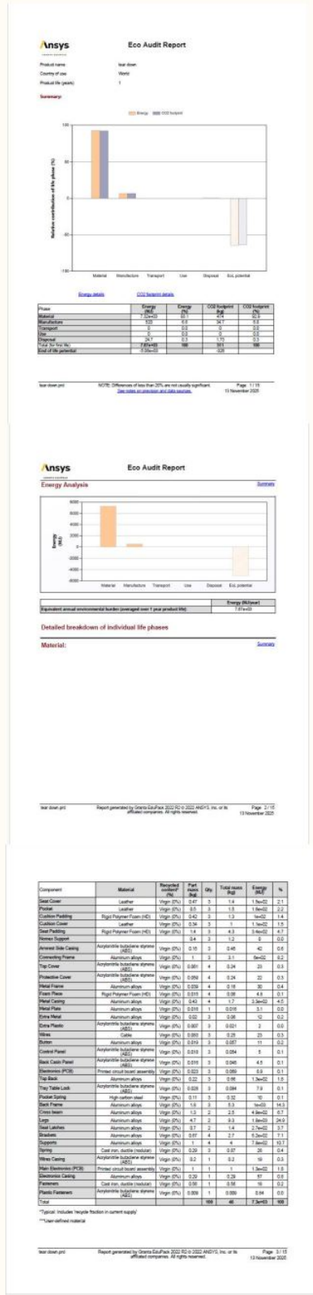


Ansys Eco Audit Report

Energy Analysis

Detailed breakdown of individual life phases

Material:



Ansys Eco Audit Report

Energy Analysis

Detailed breakdown of individual life phases

Material:

Cushioning Disassembly Comparison

Doy's Ultra Slim Seat



Tear Down Seat



1. Removing the Bottom Cushion

The patented design enables effortless removal of the cushions.





Cushions are removed by unclipping bottom buttons, making removal difficult.

2. Removing seat covers



Because it covers only the front, the Velcro design allows for quick removal.

Velcro-attached covers on both front and back make removal slower.



3. Removing the Seat Padding

The seat padding is held in place by the cushion and a single Velcro attachment for easy removal.





The seat padding is secured by a back tab and a single Velcro attachment.

4. Disassembled Seats



DOY's design allows much faster and easier access to the metal parts than the teardown design.

